PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

CogniSense: ML–Based Cognitive Load and Fatigue Analytics

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Project Guide
Ms. Aavani N.

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

- **SDG 3 – Good Health and Well-Being** 💗🧠
  - Predicts cognitive overload and mental fatigue using machine learning
  - Enables early burnout detection and preventive mental health support
  - Promotes healthier academic and workplace environments
- **SDG 4 – Quality Education** 📚🎓
  - Analyzes student cognitive load to improve learning effectiveness
  - Supports balanced academic schedules and reduces learning stress
  - Enhances student engagement and academic performance
- **SDG 11 – Sustainable Cities and Communities** 🏙️🤝
  - Supports healthier institutions and workplaces within communities
  - Encourages sustainable human resource planning and well-being

Introduction

- Increasing productivity demands in academic and workplace environments lead to high cognitive workload, mental fatigue, and burnout.
- Existing systems focus mainly on output metrics and fail to measure actual cognitive and mental load.
- **CogniSense** is an AIML-driven system that uses machine learning to analyze and predict cognitive workload and fatigue.
- By integrating behavioral data and time-series analysis, CogniSense provides early warnings, visual insights, and recommendations to promote balanced and sustainable productivity.

Introduction

- **Objectives**

- **To model cognitive workload and forecast burnout risks** using time-series behavioral insights using LSTM.

- **To automate "down-time" alerts and recovery planning** when high mental fatigue is detected using Bayesian Networks.

- **To generate privacy-preserving heatmaps** that visualize team fatigue clusters for data-driven decision-making using K-Means Clustering.

- **To adjust task distribution and meeting timings** based on predicted peak cognitive hours using Genetic Algorithms.

Literature Survey of the existing system (Most Relevant to Least)

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Our Evaluations
1.	**Classifying Cognitive Workload Using Machine Learning Techniques and Non-Intrusive Wearable Devices [1]**	Y. Liu, N. S. Grimaldi, et al.	**2024**	Successfully classified different levels of cognitive load using consumer-grade wearables.	Utilizes Machine Learning (ML) models on data collected from non-intrusive wearable sensors.	Demonstrates high practical utility for real-world monitoring, though sensor precision may vary compared to clinical tools.
2.	**Granular Estimation of User Cognitive Workload Using Multi-Modal Physiological Sensors [2]**	J. Wang, C. Stevens, et al.	**2024**	Achieved a detailed "granular" estimation of workload levels rather than binary classification.	Employs multi-modal physiological sensor fusion combined with advanced estimation algorithms.	Provides high resolution in workload detection, but multi-sensor setups can be cumbersome for the user.

Literature Survey of the existing system (Most Relevant to Least)

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Our Evaluations
3.	**Medical Intelligence Using PPG Signals and Hybrid Learning at the Edge to Detect Fatigue [3]**	P. Liu, Y. Song, et al.	**2024**	Developed a system to detect physical fatigue in real-time using localized (Edge) computing.	Uses Photoplethysmogram (PPG) signals and a Hybrid Learning architecture implemented on Edge devices.	Efficient for real-time applications and privacy-preserving, though PPG can be sensitive to motion artifacts.
4.	**Predicting Cognitive Scores from Wearable-Based Digital Physiological Features Using ML [4]**	Y. G. Rykov, M. D. Patterson, et al.	**2024**	Linked physiological digital biomarkers directly to cognitive performance scores.	Machine Learning models trained on digital physiological features extracted from wearable devices.	Strong clinical relevance for health monitoring, but requires large longitudinal datasets for high reliability.

Literature Survey of the existing system (Most Relevant to Least)

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Our Evaluations
5.	**Detecting and Predicting Pilot Mental Workload Using Heart Rate Variability [5]**	P. Wang, R. Houghton, et al.	**2024**	Identified HRV as a reliable metric for predicting mental workload in high-stakes aviation.	A Systematic Review of studies using Heart Rate Variability (HRV) and predictive modeling.	Comprehensive overview of the field, highlighting that HRV is effective but influenced by physical stress.
6.	**EEG-Based Mental Workload Prediction Using Deep Learning for AR Systems [6]**	X. Zhang, Y. Chen, et al.	**2023**	Improved workload prediction accuracy within Augmented Reality (AR) environments.	Implements Deep Learning models (CNN/RNN) on Electroencephal ogram (EEG) data.	Achieves high sensitivity to mental state changes, but EEG hardware can be intrusive for AR users.

Literature Survey of the existing system (Most Relevant to Least)

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Our Evaluations
7.	**ML Performance in EEG-Based Mental Workload Classification Across Task Types [7]**	M. Pušica, B. Mijović, et al.	**2025**	Evaluated how different task types affect the accuracy of EEG-based ML classifiers.	A Systematic Review focusing on EEG classification performance across varied task domains.	Highlights the lack of "one-size-fits-all" models; performance is highly dependent on the specific task type.
8.	**Novel Multimodal Sensing and ML Strategies to Classify Workload in Laparoscopic Surgery [8]**	R. Naik, A. Rubio-Solis, et al.	**2025**	Enhanced the safety of surgical procedures by monitoring surgeon cognitive load.	Uses Multimodal sensing (eye-tracking, heart rate, etc.) and specialized ML classification strategies.	Highly specialized and effective for high-precision tasks, but integration into the OR is technically complex.

Limitations of existing systems

- **Reactive Fatigue Detection:**
  Most current systems identify stress or burnout only after noticeable performance decline, not at early stages.

- **Lack of Cognitive Load Measurement:**
  Existing productivity and monitoring tools focus on task completion and time tracking, ignoring mental workload and cognitive strain.

- **Absence of Predictive Analytics:**
  Few systems use machine learning or time-series forecasting to predict future fatigue or burnout risks.

- **Generic and Non-Personalized Models:**
  Current solutions apply uniform thresholds and rules, failing to adapt to individual work patterns and stress tolerance.

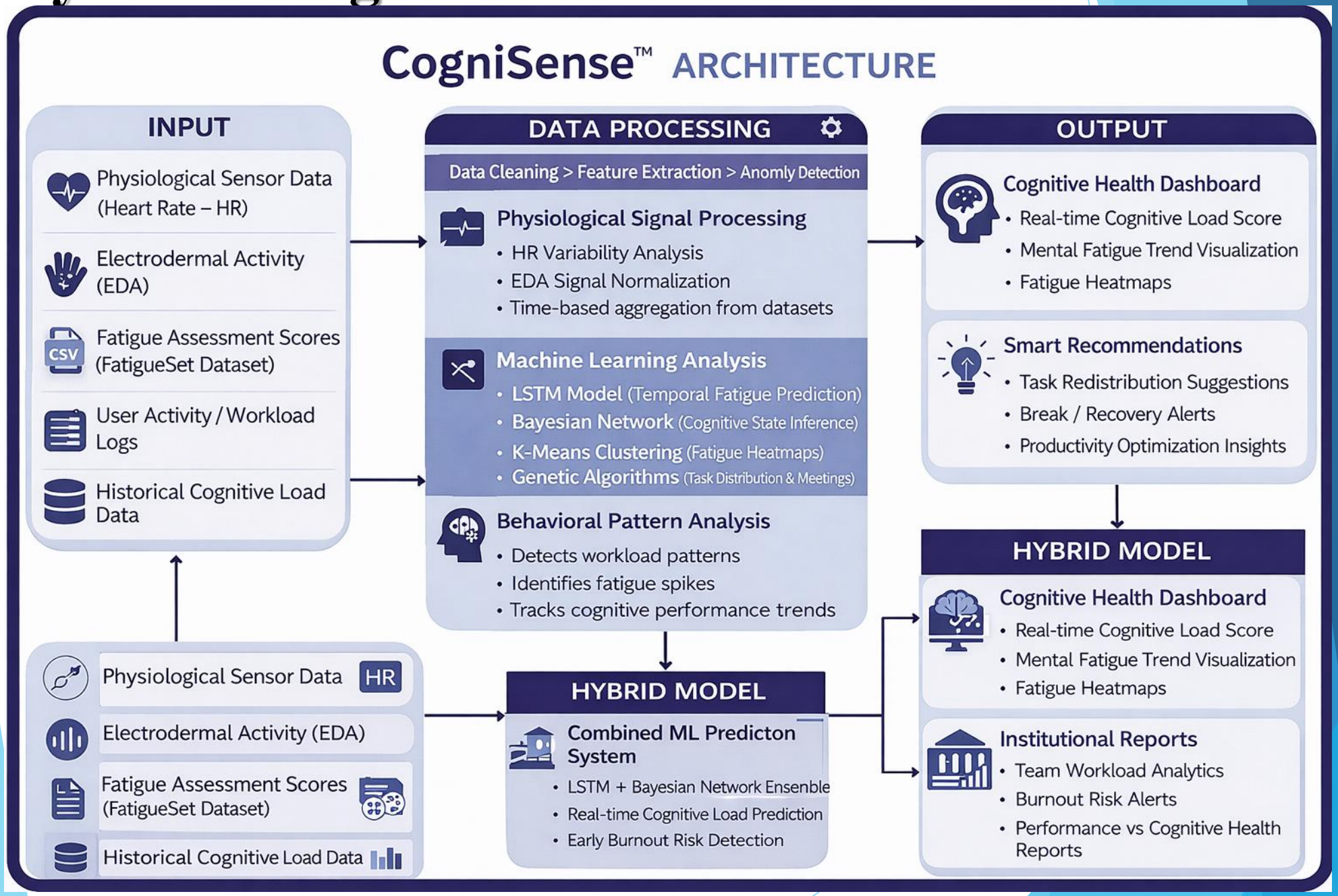
Problem statement

- **The Challenge:** Cognitive overload, mental fatigue, and burnout are increasingly common in academic and workplace environments due to continuous workloads, multitasking, and tight schedules. Existing systems mainly focus on productivity outcomes and fail to monitor real-time cognitive load or detect early signs of fatigue.

- Our Goal is to Design an Intelligent and Ethical Solution that:

  - Uses machine learning models to analyze cognitive workload and mental fatigue patterns.
  - Predicts early burnout risks before noticeable performance decline.
  - Considers individual cognitive differences for more accurate assessment.
  - Ensures privacy-aware and ethical monitoring of mental workload.
  - Provides insights and alerts to support balanced productivity and well-being.

System Design



Technologies

- **Frontend Development:**
  - React.js (Vite), HTML, TypeScript, CSS, Tailwind CSS → Interactive cognitive health dashboard
  - Radix UI, Lucide Icons → UI components & icons
  - Chart.js → Visualization of cognitive load and fatigue metrics

- **Backend Development:**
  - Python (FastAPI) → ML analysis engine and APIs
  - Node.js (Express) → Algorithms instance & comparative analysis
  - Signal Processing → Butterworth Filters, Hjorth Parameters for fatigue detection

- **Database:**
  - Physiological sensor datasets (EDA, HR, ECG, ACC)
  - FatigueSet multi-subject dataset architecture
  - Approx combined datasets size – [39,62,100 x 12]

Technologies

- **Machine Learning Models:**
  - LSTM (Cognitive Workload & Burnout Prediction)
  - Bayesian Network (Mental Fatigue Modeling)
  - K-Means Clustering (Fatigue Heatmaps)
  - Genetic Algorithm (Task Distribution & Meeting Optimization)

- **Data Sources:**
  - Physiological sensor logs (HR, EDA, ECG, ACC)
  - Fatigue assessment datasets
  - Workload activity logs
  - Avg. Environmental context (common reports)

Methodologies

1. **Data Collection:** Collect academic and workplace activity data including task duration, work/study hours, deadlines, meetings, breaks, and optional self-reported fatigue inputs via secure forms, logs, or APIs

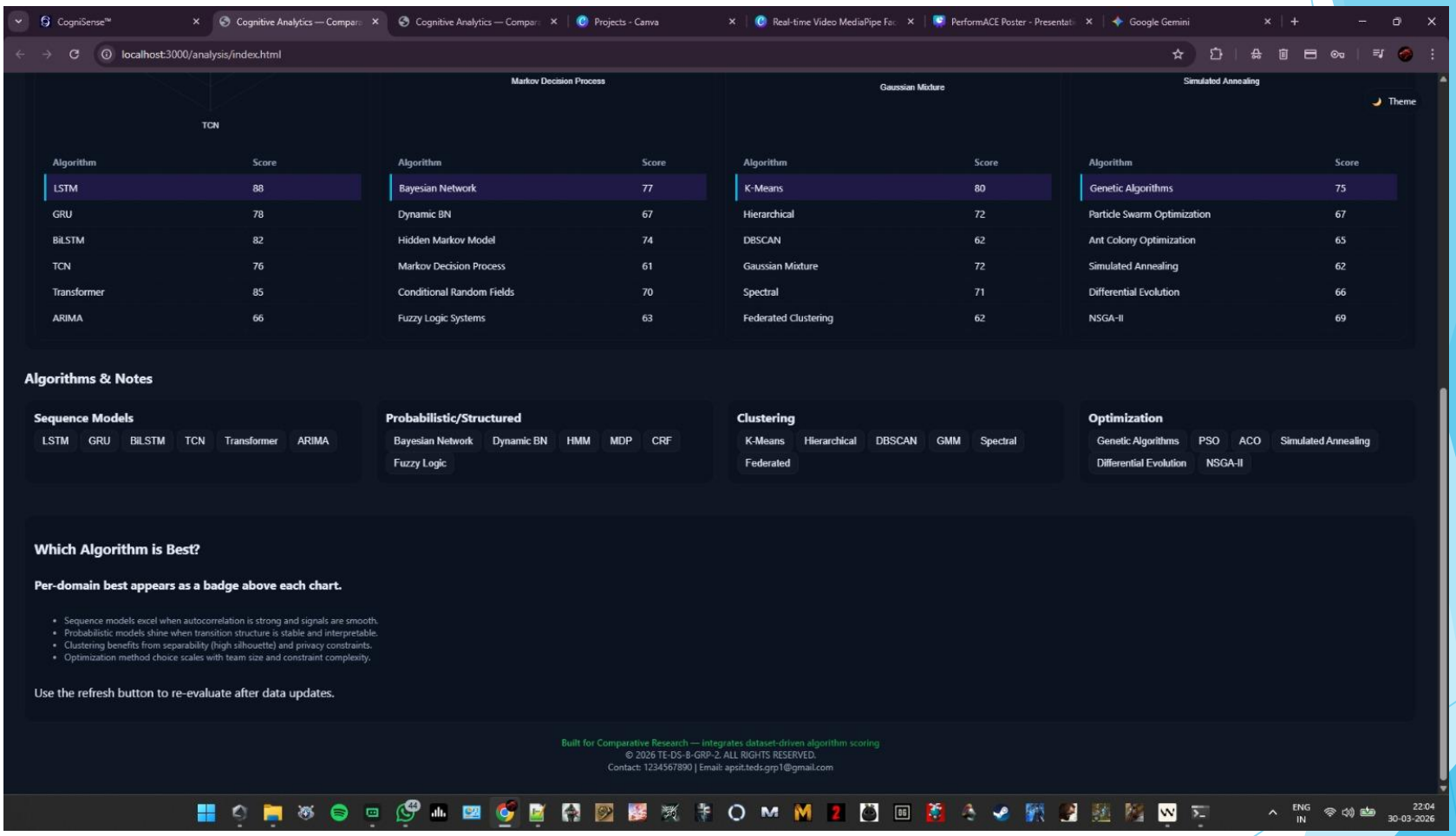
2. **Data Preprocessing:** Clean missing and noisy data, normalize workload metrics, and apply anonymization to ensure privacy protection.

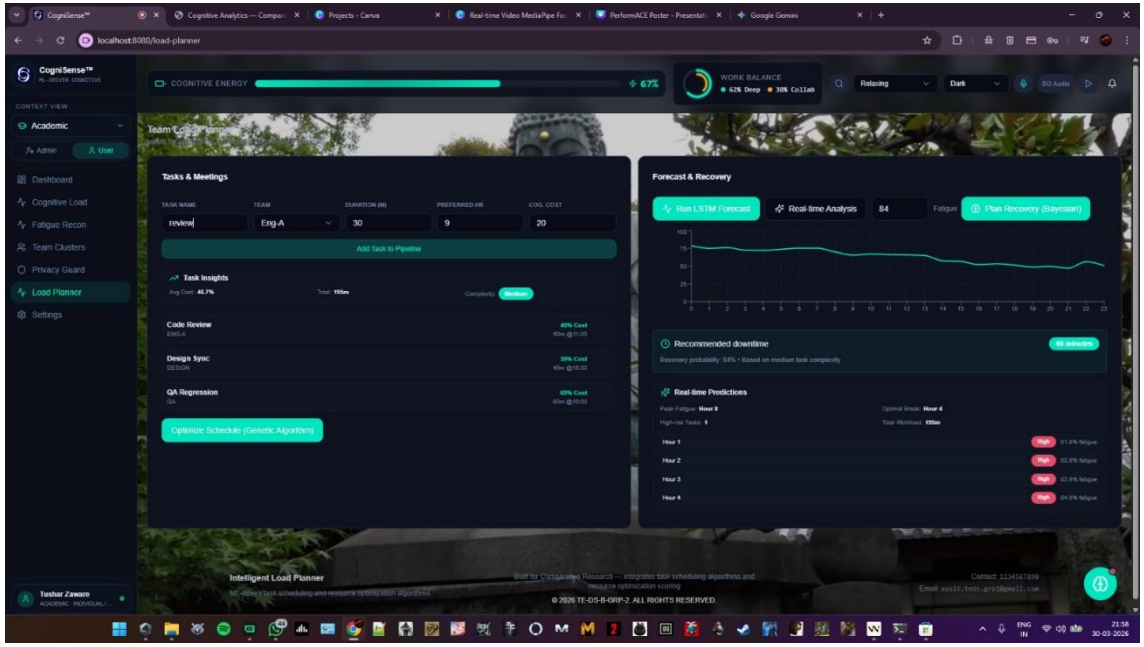
3. **Feature Engineering:** Extract features such as task intensity, workload duration, break frequency, and temporal patterns to model cognitive load trends.

4. **Model Training (Python ML Models):** Train ML models including – Linear Regression, LSTM, K-Means Clustering, and Bayesian Models for fatigue and burnout analysis

5. **Output & Visualization:** Predict cognitive fatigue and burnout risks in advance and identify high-risk workload periods using predictive analytics

Implementation

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ML-DRIVEN COGNITIVE
COGNITIVE ENERGY
67%
WORK BALANCE
62% Deep
38% Collab
Relaxing
Dark
BG Audio
CONTEXT VIEW
Academic
Admin
User
Dashboard
Cognitive Load
Fatigue Recon
Team Clusters
Privacy Guard
Load Planner
Settings
Tasks & Meetings
TASK NAME
TEAM
DURATION (M)
PREFERRED HR
COG. COST
review
Eng-A
30
9
20
Add Task to Pipeline
Task Insights
Code Review
Design Sync
QA Regression
Optimize Schedule (Genetic Algorithm)
Forecast & Recovery
Run LSTM Forecast
Real-time Analysis
84
Fatigue
Plan Recovery (Bayesian)
Recommended downtime
Real-time Predictions
Hour 1
Hour 2
Hour 3
Hour 4
Intelligent Load Planner
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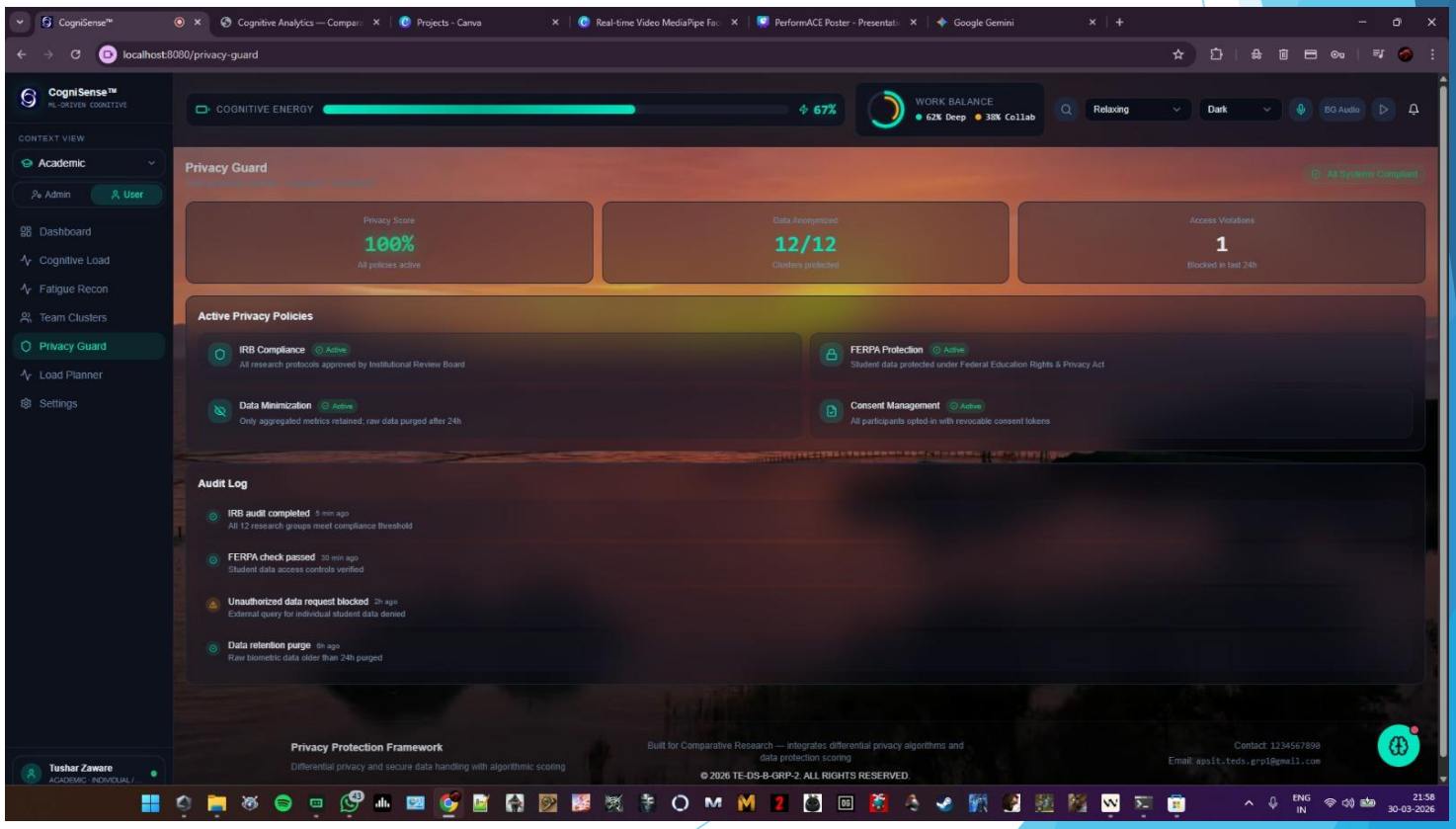
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Settings
Privacy Guard
All Systems Compliant
Privacy Score
100%
All policies active
Data Anonymized
12/12
Clusters protected
Access Violations
1
Blocked in last 24h
Active Privacy Policies
IRB Compliance
Active
All research protocols approved by Institutional Review Board
FERPA Protection
Active
Student data protected under Federal Education Rights & Privacy Act
Data Minimization
Active
Only aggregated metrics retained; raw data purged after 24h
Consent Management
Active
All participants opted-in with revocable consent tokens
Audit Log
IRB audit completed
5 min ago
All 12 research groups meet compliance threshold
FERPA check passed
30 min ago
Student data access controls verified
Unauthorized data request blocked
2h ago
External query for individual student data denied
Data retention purge
6h ago
Raw biometric data older than 24h purged
Privacy Protection Framework
Differential privacy and secure data handling with algorithmic scoring
Built for Comparative Research — integrates differential privacy algorithms and data protection scoring
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Email: apsit.teds.grp1@gmail.com
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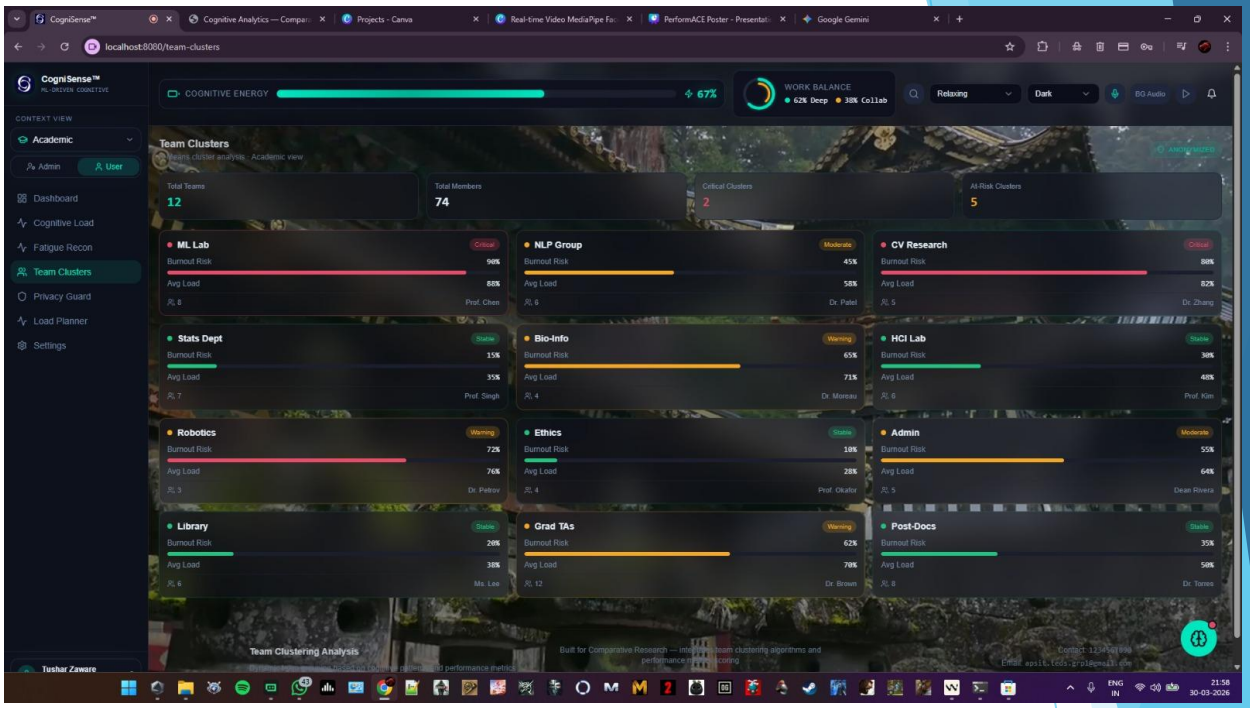
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Dashboard
Cognitive Load
Fatigue Recon
Team Clusters
Privacy Guard
Load Planner
Settings
Team Clusters
Total Teams
12
Total Members
74
Critical Clusters
2
At-Risk Clusters
5
ML Lab
NLP Group
CV Research
Stats Dept
Bio-Info
HCI Lab
Robotics
Ethics
Admin
Library
Grad TAs
Post-Docs
Burnout Risk
Avg Load
Prof. Chen
Dr. Patel
Dr. Zhang
Prof. Singh
Dr. Moreau
Prof. Kim
Dr. Petrov
Prof. Okafor
Dean Rivera
Ms. Lee
Dr. Brown
Dr. Torres
Team Clustering Analysis
Tushar Zaware

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Cognitive Load Analysis
Real-time cognitive load monitoring · Academic view
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Export
Live: 0%
Live
0%
Avg. Cognitive Load
85%
Peak Load Today
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Decay Rate (λ)
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Recovery ETA
Hourly Cognitive Load
Cognitive Profile
Fatigue Predictions
ML-based mental fatigue forecasts from fatigueset
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Cognitive Load Decomposition
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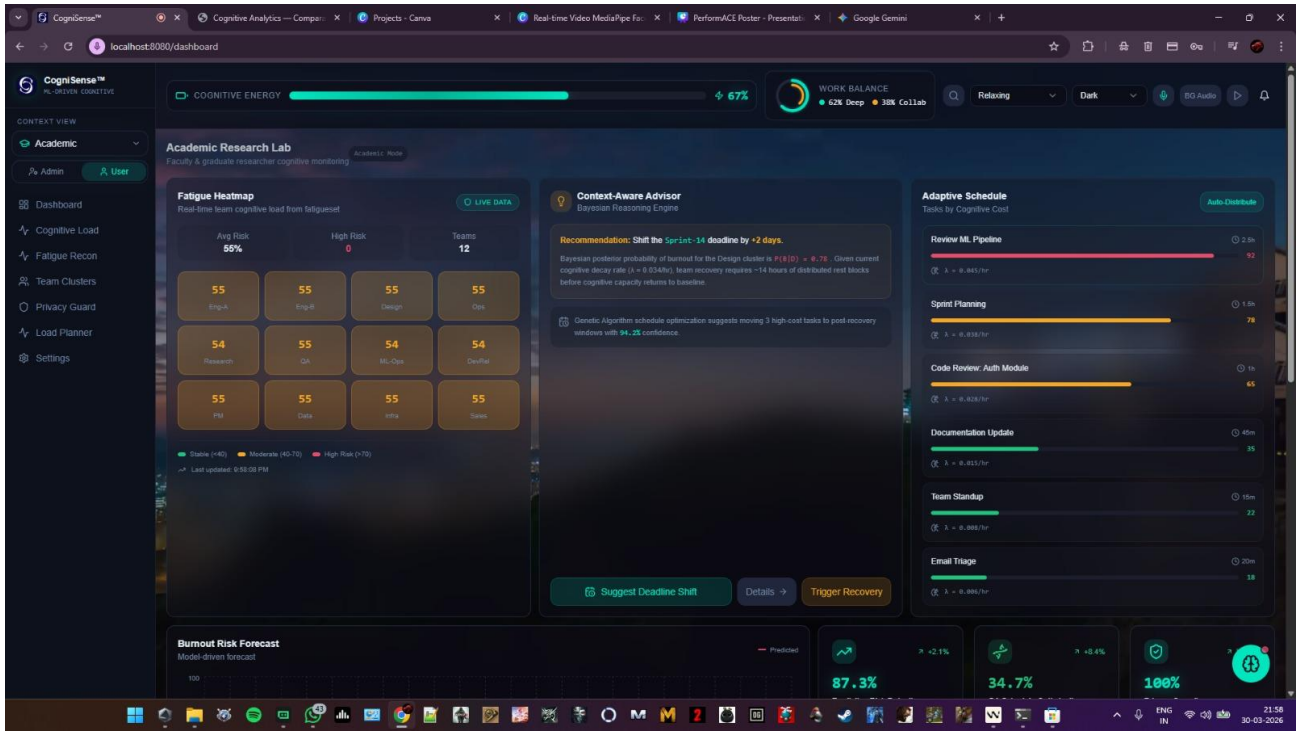
Intrinsic
Extraneous
Thesis Review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Fatigue Recon
Team Clusters
Privacy Guard
Load Planner
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Academic Research Lab
Faculty & graduate researcher cognitive monitoring
Academic Mode
Fatigue Heatmap
Real-time team cognitive load from fatigueset
LIVE DATA
Avg Risk
55%
High Risk
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55
Eng-A
55
Eng-B
55
Design
55
Ops
54
Research
55
QA
54
ML-Ops
54
DevRel
55
PM
55
Data
55
Infra
55
Sales
Stable (<40)
Moderate (40-70)
High Risk (>70)
Last updated: 0:58:08 PM
Context-Aware Advisor
Bayesian Reasoning Engine
Recommendation: Shift the Sprint-14 deadline by +2 days.
Bayesian posterior probability of burnout for the Design cluster is P(B|D) = 0.78 . Given current cognitive decay rate (λ = 0.034/hr) team recovery requires ~14 hours of distributed rest blocks before cognitive capacity returns to baseline.
Genetic Algorithm schedule optimization suggests moving 3 high-cost tasks to post-recovery windows with 94.2% confidence.
Suggest Deadline Shift
Details →
Trigger Recovery
Adaptive Schedule
Tasks by Cognitive Cost
Auto-Distribute
Review ML Pipeline
2.5h
92
λ = 0.045/hr
Sprint Planning
1.5h
78
λ = 0.038/hr
Code Review: Auth Module
1h
65
λ = 0.028/hr
Documentation Update
45m
35
λ = 0.015/hr
Team Standup
15m
22
λ = 0.008/hr
Email Triage
20m
18
λ = 0.006/hr
Burnout Risk Forecast
Model-driven forecast
Predicted
87.3%
34.7%
100%
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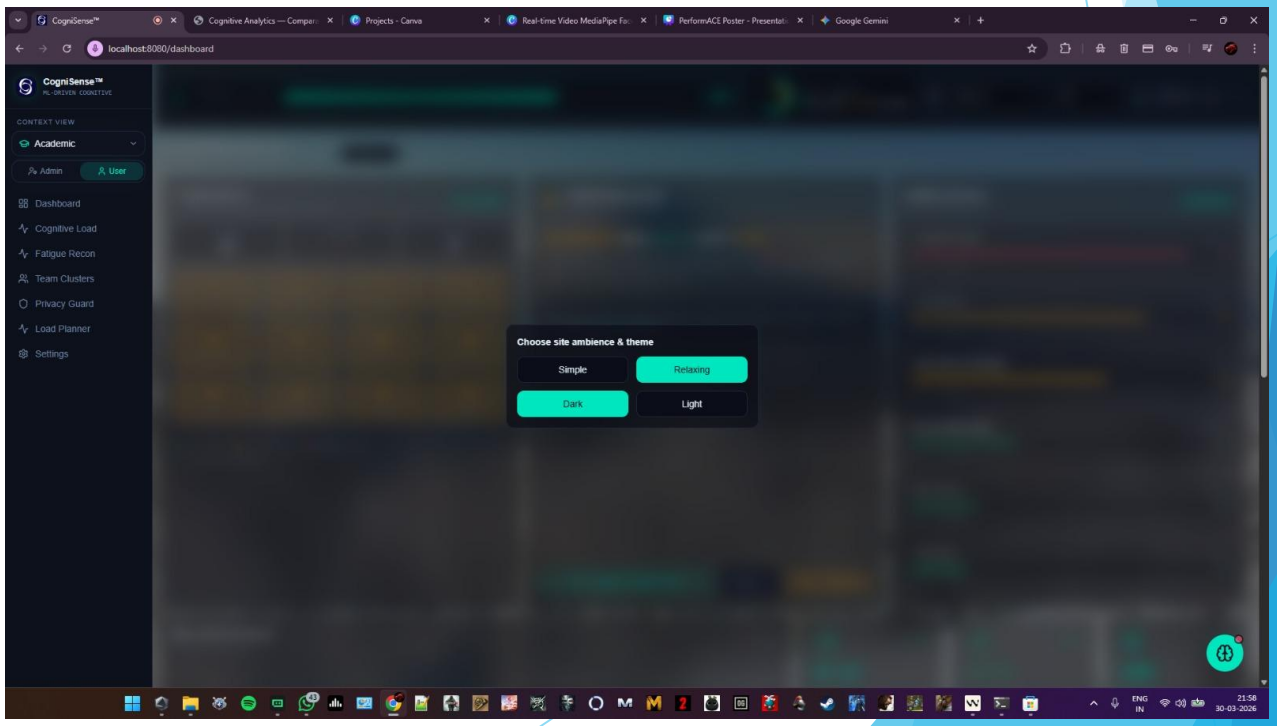
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ML-DRIVEN COGNITIVE
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Team Clusters
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Load Planner
Settings
Choose site ambience & theme
Simple
Relaxing
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30-03-2026

Conclusion

- Successfully developed an **end-to-end ML-based system** for **collecting, preprocessing, and analyzing cognitive workload data** from academic and workplace environments.
- **CogniSense** predicts mental fatigue and burnout early using **workload modeling** and **time-series behavioral analysis**.
- Provides **data-driven insights** and **interactive dashboards** to identify **cognitive load trends and high-risk periods**.
- Supports balanced productivity and improved mental well-being in academic and workplace settings.
- Demonstrates the application of **ethical and privacy-aware AI for human-centric analytics**.
- Future scope includes wearable integration, real-time fatigue detection, adaptive scheduling, and advanced deep learning models.

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URL: https://pubmed.ncbi.nlm.nih.gov/39482204/

Thank You...!!